

Since many of you found this confusing the last time, let us try to redo it

We consider a multivariable function f , which means that it takes two arguments because its domain is now $\mathbb{R}^2$

instead of $\mathbb{R}$

Multivariable calculus

What is the derivative of

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}$$

○

We are interested in the derivative of f which we have seen is not as straightforward to define

Multivariable calculus

What is the derivative of

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}$$



What is the derivative of

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}$$

along a parametrization $\theta : (a, b) \rightarrow \mathbb{R}^2$?

so we refine the question and ask what the derivative of f is *along a curve*

What is the derivative of

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}$$

along a parametrization $\theta : (a, b) \rightarrow \mathbb{R}^2$?

The function along a curve parametrized by θ is simply $f(\theta(t))$

Multivariable calculus

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Multivariable calculus

What is the derivative of

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along a parametrization $\theta : (a, b) \rightarrow \mathbb{R}^2$?

$$\frac{d}{dt} f(\theta(t))$$



What is the derivative of

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}$$

along a parametrization $\theta : (a, b) \rightarrow \mathbb{R}^2$?

$$\frac{d}{dt} f(\theta(t))$$

So this is undoubtedly the definition of the derivative of f along the parametrization θ

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$$\frac{d}{dt} f(\theta(t))$$

Chain rule



Multivariable calculus

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$$\frac{d}{dt} f(\theta(t))$$

Chain rule (a computational device)

Now one should initially view the chain rule as strictly a computational device

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$$\frac{d}{dt} f(\theta(t))$$

Chain rule (a computational device)

$$\frac{d}{dt} f(\theta(t)) = \nabla f \cdot \theta'(t)$$

but it leads us to some interesting conclusions

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Chain rule (a computational device)

$$\frac{d}{dt} f(\theta(t)) = \nabla f \cdot \theta'(t)$$

by splitting the computation of the derivative along the curve in two parts

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one part is the velocity of θ

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$$\frac{d}{dt} f(\theta(t))$$

Chain rule (a computational device)

$$\frac{d}{dt} f(\theta(t)) = \nabla f \cdot \theta'(t)$$

$$\text{where } \nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right)$$

while the other is called the grad of f and denoted ∇f

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where $\nabla f = (\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y})$ where $\frac{\partial f}{\partial x}$ is obtained by keeping y fixed and taking the derivative with respect to x

and is obtained by introducing this new computational device called the partial derivatives

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To understand all this let us consider this example

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Example

$$f(x, y) = x^2 + y^2$$

where we define a function that is geometrically the the square of the distance from the origin

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Example

$$f(x, y) = x^2 + y^2$$

$$\theta(t) = (\cos(t), \sin(t)) \quad t \in (-\pi, \pi)$$

We will find its rate of change along a circle with center at the origin

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Example

$$f(x, y) = x^2 + y^2$$

$$\theta(t) = (\cos(t), \sin(t)) \quad t \in (-\pi, \pi)$$

$$\frac{d}{dt} \Big|_{t=0} f(\theta(t))$$

Notice that along the circle, since its center is the origin, the square of the distance will be constant

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Example

$$f(x, y) = x^2 + y^2$$

$$\theta(t) = (\cos(t), \sin(t)) \quad t \in (-\pi, \pi)$$

$$\frac{d}{dt} \big|_{t=0} f(\theta(t)) = 0$$

and, therefore, the derivative will be 0

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$$f(x, y) = x^2 + y^2$$

$$\theta(t) = (\cos(t), \sin(t)) \quad t \in (-\pi, \pi)$$

$\frac{d}{dt}|_{t=0} f(\theta(t)) = 0$ because along the curve defined θ , f is

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Let us use the chain rule instead

Now let us get used to the chain rule by applying it here

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Example

$$f(x, y) = x^2 + y^2$$

$$\theta(t) = (\cos(t), \sin(t)) \quad t \in (-\pi, \pi)$$

$\frac{d}{dt}|_{t=0} f(\theta(t)) = 0$ because along the curve defined θ , f is constant

Let us use the chain rule instead

$$\frac{\partial f}{\partial x}$$

So let us consider the partial derivative with respect to x

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Example

$$f(x, y) = x^2 + y^2$$

$$\theta(t) = (\cos(t), \sin(t)) \quad t \in (-\pi, \pi)$$

$\frac{d}{dt} f(\theta(t))|_{t=0} = 0$ because along the curve defined θ , f is constant

Let us use the chain rule instead

$$\frac{\partial f}{\partial x} \Big|_{x=1}$$

And we will do it at $x = 1$ because that is the x -coordinate of $\theta(0)$ and we are doing this at $t = 0$

Multivariable calculus

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Example

$$f(x, y) = x^2 + y^2$$

$$\theta(t) = (\cos(t), \sin(t)) \quad t \in (-\pi, \pi)$$

$\frac{d}{dt} f(\theta(t)) = 0$ because along the curve defined θ , f is constant

Let us use the chain rule instead

$\frac{\partial f}{\partial x} \big|_{x=1}$ which means, keep $y = 0$ fixed, so that it becomes a function of only x

For this reason, we have to keep $y = 0$ because that is the y -coordinate of $\theta(0)$

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Example

$$f(x, y) = x^2 + y^2$$

$$\theta(t) = (\cos(t), \sin(t)) \quad t \in (-\pi, \pi)$$

$\frac{d}{dt} f(\theta(t)) = 0$ because along the curve defined θ , f is constant

Let us use the chain rule instead

$\frac{\partial f}{\partial x} \big|_{x=1}$ which means, keep $y = 0$ fixed, so that it becomes a function of only x and then take the derivative at $x = 1$

and then by definition of the newly introduced partial derivatives,

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$\frac{\partial f}{\partial x} \big|_{x=1}$ which means, keep $y = 0$ fixed, so that it becomes a function of only x and then take the derivative at $x = 1$

now that we have fixed the y , we have to now simply take the derivative

Multivariable calculus

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Example

$$f(x, y) = x^2 + y^2$$

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$\frac{d}{dt}|_{t=0} f(\theta(t)) = 0$ because along the curve defined θ , f is constant

Let us use the chain rule instead

$\frac{\partial f}{\partial x}|_{x=1}$ which means, keep $y = 0$ fixed, so that it becomes a function of only x and then take the derivative at $x = 1$

$$\frac{\partial f}{\partial x}|_{x=1} = 2x|_{x=1}$$

And that is why we get this computation

Multivariable calculus

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Example

$$f(x, y) = x^2 + y^2$$

$$\theta(t) = (\cos(t), \sin(t)) \quad t \in (-\pi, \pi)$$

$\frac{d}{dt}|_{t=0} f(\theta(t)) = 0$ because along the curve defined θ , f is constant

Let us use the chain rule instead

$\frac{\partial f}{\partial x}|_{x=1}$ which means, keep $y = 0$ fixed, so that it becomes a function of only x and then take the derivative at $x = 1$

$$\frac{\partial f}{\partial x}|_{x=1} = 2x|_{x=1} = 2$$

and obtain the partial derivative with respect to x as 2

Multivariable calculus

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Example

$$f(x, y) = x^2 + y^2$$

$$\theta(t) = (\cos(t), \sin(t)) \quad t \in (-\pi, \pi)$$

$\frac{d}{dt}|_{t=0} f(\theta(t)) = 0$ because along the curve defined θ , f is constant

Let us use the chain rule instead

$\frac{\partial f}{\partial x}|_{x=1}$ which means, keep $y = 0$ fixed, so that it becomes a function of only x and then take the derivative at $x = 1$

$$\frac{\partial f}{\partial x}|_{x=1} = 2x|_{x=1} = 2$$

Similarly,

$$\frac{\partial f}{\partial y}|_{y=0} = 0$$

It should be obvious that the partial derivative with respect to y is 0

Multivariable calculus

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Example

$$f(x, y) = x^2 + y^2$$

$$\theta(t) = (\cos(t), \sin(t)) \quad t \in (-\pi, \pi)$$

$\frac{d}{dt}|_{t=0} f(\theta(t)) = 0$ because along the curve defined θ , f is constant

Let us use the chain rule instead

$\frac{\partial f}{\partial x}|_{x=1}$ which means, keep $y = 0$ fixed, so that it becomes a function of only x and then take the derivative at $x = 1$

$$\frac{\partial f}{\partial x}|_{x=1} = 2x|_{x=1} = 2$$

Similarly,

$$\frac{\partial f}{\partial y}|_{y=0} = 0$$

$$\nabla f = (2, 0)$$

and therefore the grad is the vector $(2, 0)$

Multivariable calculus

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$$\frac{d}{dt} f(\theta(t))$$

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where $\nabla f = (\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y})$ where $\frac{\partial f}{\partial x}$ is obtained by keeping y fixed and taking the derivative with respect to x

Example

$$f(x, y) = x^2 + y^2$$

$$\theta(t) = (\cos(t), \sin(t)) \quad t \in (-\pi, \pi)$$

$\frac{d}{dt}|_{t=0} f(\theta(t)) = 0$ because along the curve defined θ , f is constant

Let us use the chain rule instead

$\frac{\partial f}{\partial x}|_{x=1}$ which means, keep $y = 0$ fixed, so that it becomes a function of only x and then take the derivative at $x = 1$

$$\frac{\partial f}{\partial x}|_{x=1} = 2x|_{x=1} = 2$$

Similarly,

$$\frac{\partial f}{\partial y}|_{y=0} = 0$$

$$\nabla f = (2, 0)$$

$$\theta'(0) = (-\sin(0), \cos(0)) = (0, 1)$$

This computation of the velocity is a standard one

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Example

$$f(x, y) = x^2 + y^2$$

$$\theta(t) = (\cos(t), \sin(t)) \quad t \in (-\pi, \pi)$$

$\frac{d}{dt}|_{t=0} f(\theta(t)) = 0$ because along the curve defined θ , f is constant

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Similarly,

$$\frac{\partial f}{\partial y}|_{y=0} = 0$$

$$\nabla f = (2, 0)$$

$$\theta'(0) = (-\sin(0), \cos(0)) = (0, 1)$$

$$\nabla(f) \cdot \theta'(0) = (2, 0) \cdot (0, 1) = 0$$

we can see that chain rule also tells us that the derivative, which is the dot product shown here, is 0

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$$\nabla f = (2, 0)$$

$$\theta'(0) = (-\sin(0), \cos(0)) = (0, 1)$$

$$\nabla f \cdot \theta'(0) = (2, 0) \cdot (0, 1) = 0$$

Why is the direction of ∇f in the direction of the steepest increase of f ?

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$$\theta'(0) = (-\sin(0), \cos(0)) = (0, 1)$$

$$\nabla(f) \cdot \theta'(0) = (2, 0) \cdot (0, 1) = 0$$

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Observations

Example

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$$\theta'(0) = (-\sin(0), \cos(0)) = (0, 1)$$

$$\nabla f \cdot \theta'(0) = (2, 0) \cdot (0, 1) = 0$$

Let us make a few observations about the nature of the chain rule

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Chain rule (a computational device)

$$\frac{d}{dt} f(\theta(t)) = \nabla f \cdot \theta'(t)$$

where $\nabla f = (\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y})$ where $\frac{\partial f}{\partial x}$ is obtained by keeping y fixed and taking the derivative with respect to x

Observations

1. ∇f does not depend on θ

Example

$$f(x, y) = x^2 + y^2$$

$$\theta(t) = (\cos(t), \sin(t)) \quad t \in (-\pi, \pi)$$

$\frac{d}{dt}|_{t=0} f(\theta(t)) = 0$ because along the curve defined θ , f is constant

Let us use the chain rule instead

$\frac{\partial f}{\partial x}|_{x=1}$ which means, keep $y = 0$ fixed, so that it becomes a function of only x and then take the derivative at $x = 1$

$$\frac{\partial f}{\partial x}|_{x=1} = 2x|_{x=1} = 2$$

Similarly,

$$\frac{\partial f}{\partial y}|_{y=0} = 0$$

$$\nabla f = (2, 0)$$

$$\theta'(0) = (-\sin(0), \cos(0)) = (0, 1)$$

$$\nabla f \cdot \theta'(0) = (2, 0) \cdot (0, 1) = 0$$

Notice that the ∇f in the right hand side of the highlighted formula does not involve θ

Multivariable calculus

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Similarly, the right hand side has no f in the $\theta'(t)$ part

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These two observations naturally lead us to a new concept of directional derivatives

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which is originating at the point p at which we are taking the derivative

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But we do it by using the easier concept of the derivative of f along the parametrization

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by choosing a *any* curve passing through p

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but parametrized so that its velocity is the given vector $\mathbf{v}$

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and now we just find out the derivative along that curve

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which owing to the highlighted observation, does not really depend on the θ

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any other parametrization would have given the same answer

Multivariable calculus

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$$f : \mathbb{R}^2 \rightarrow \mathbb{R}$$

along a parametrization $\theta : (a, b) \rightarrow \mathbb{R}^2$?

$$\frac{d}{dt} f(\theta(t))$$

Chain rule (a computational device)

$$\frac{d}{dt} f(\theta(t)) = \nabla f \cdot \theta'(t)$$

where $\nabla f = (\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y})$ where $\frac{\partial f}{\partial x}$ is obtained by keeping y fixed and taking the derivative with respect to x

Observations

1. ∇f does not depend on θ
2. $\theta'(t)$ does not depend on f

Directional derivative

We can define the derivative of f in the direction of $\mathbf{v}$ originating at the point p

Choose any θ_0 so that $\theta_0(t_0) = p$

Example

$$f(x, y) = x^2 + y^2$$

$$\theta(t) = (\cos(t), \sin(t)) \quad t \in (-\pi, \pi)$$

$\frac{d}{dt}|_{t=0} f(\theta(t)) = 0$ because along the curve defined θ , f is constant

Let us use the chain rule instead

$\frac{\partial f}{\partial x}|_{x=1}$ which means, keep $y = 0$ fixed, so that it becomes a function of only x and then take the derivative at $x = 1$

$$\frac{\partial f}{\partial x}|_{x=1} = 2x|_{x=1} = 2$$

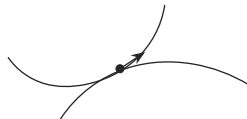
Similarly,

$$\frac{\partial f}{\partial y}|_{y=0} = 0$$

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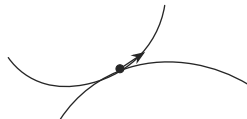
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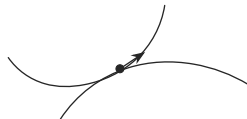
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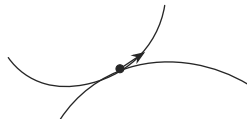
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demonstrating that even though we use the parametrization to find how it f is varying along it

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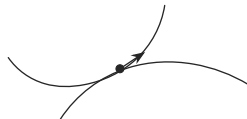
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It really depends only on the velocity vector

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We will use all this to understand surfaces

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We define a surface,
 $\sigma : U \rightarrow \mathbb{R}^3$

which, remember is defined by using a “surface patch” σ

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which is simply a function that takes a two dimensional piece U and “stretches and bends it” in the 3-space

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but in order to not mess up the dimension it needs to be nice enough

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we will see later what that means

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but for one, one to one ensures it does not collapse the space into something lower dimension

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Now our main interest will be in understanding the curvature of a surface at a point

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But how do we define this?

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Why not learn from our lesson with derivatives of a multivariable function

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and start with what we know

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Curves!!

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We will study surfaces by seeing how curves on them are forced to curve

We can try and see how the curvature of the surface

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force any curve on it to curve too

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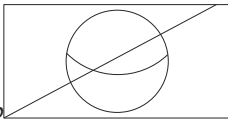
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For instance, if the surface is the sphere

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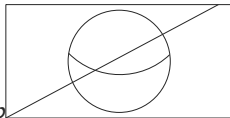
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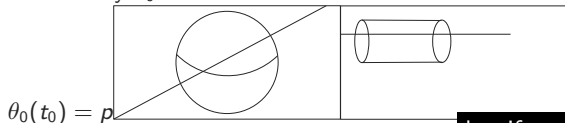
Observations

1. ∇f does not depend on θ
2. $\theta'(t)$ does not depend on f

Directional derivative

We can define the derivative of f in the direction of $\mathbf{v}$ originating at the point p

Choose any θ_0 so that



but if we have a cylinder

We define a surface,

$\sigma : U \rightarrow \mathbb{R}^3$ where $U \subset \mathbb{R}^2$ and σ is smooth, and “nice enough” like injective

We will study surfaces by seeing how curves on them are forced to curve

Multivariable calculus

What is the derivative of

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}$$

along a parametrization $\theta : (a, b) \rightarrow \mathbb{R}^2$?

$$\frac{d}{dt} f(\theta(t))$$

Chain rule (a computational device)

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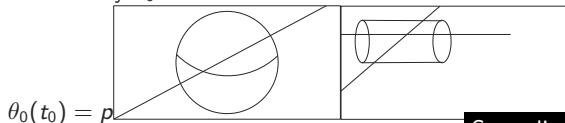
Observations

1. ∇f does not depend on θ
2. $\theta'(t)$ does not depend on f

Directional derivative

We can define the derivative of f in the direction of $\mathbf{v}$ originating at the point p

Choose any θ_0 so that



Some lines can lie on it

We define a surface,

$\sigma : U \rightarrow \mathbb{R}^3$ where $U \subset \mathbb{R}^2$ and σ is smooth, and “nice enough” like injective

We will study surfaces by seeing how curves on them are forced to curve

Multivariable calculus

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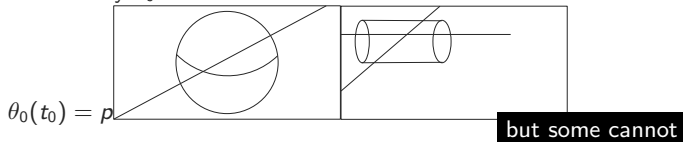
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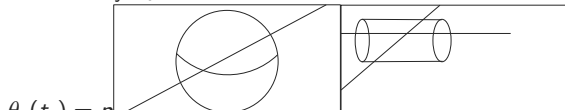
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Directional derivative

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We define a surface,

$\sigma : U \rightarrow \mathbb{R}^3$ where $U \subset \mathbb{R}^2$ and σ is smooth, and “nice enough” like injective

We will study surfaces by seeing how curves on them are forced to curve

That means, figuring out the acceleration component that keeps the particle moving along the curve on the surface

$\theta_0(t_0)$ So one way to figure out the curvature of a surface at a point is to choose a curve passing through it

Multivariable calculus

What is the derivative of

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}$$

along a parametrization $\theta : (a, b) \rightarrow \mathbb{R}^2$?

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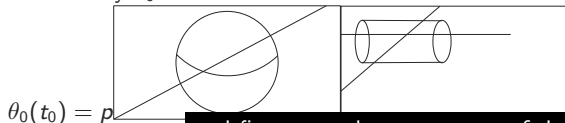
Observations

1. ∇f does not depend on θ
2. $\theta'(t)$ does not depend on f

Directional derivative

We can define the derivative of f in the direction of $\mathbf{v}$ originating at the point p

Choose any θ_0 so that



$$\theta_0(t_0) = p$$

and figure out the component of the acceleration that keeps it on the surface

We define a surface,

$\sigma : U \rightarrow \mathbb{R}^3$ where $U \subset \mathbb{R}^2$ and σ is smooth, and “nice enough” like injective

We will study surfaces by seeing how curves on them are forced to curve

That means, figuring out the acceleration component that keeps the particle moving along the curve on the surface

Multivariable calculus

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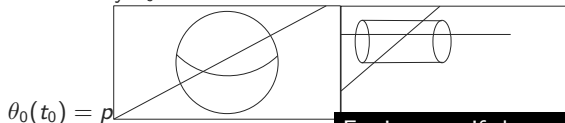
Observations

1. ∇f does not depend on θ
2. $\theta'(t)$ does not depend on f

Directional derivative

We can define the derivative of f in the direction of $\mathbf{v}$ originating at the point p

Choose any θ_0 so that



For instance, if the particle is moving on a sphere

We define a surface,

$\sigma : U \rightarrow \mathbb{R}^3$ where $U \subset \mathbb{R}^2$ and σ is smooth, and “nice enough” like injective

We will study surfaces by seeing how curves on them are forced to curve

That means, figuring out the acceleration component that keeps the particle moving along the curve on the surface

Multivariable calculus

What is the derivative of

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Chain rule (a computational device)

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where $\nabla f = (\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y})$ where $\frac{\partial f}{\partial x}$ is obtained by keeping y fixed and taking the derivative with respect to x

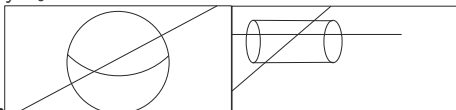
Observations

1. ∇f does not depend on θ
2. $\theta'(t)$ does not depend on f

Directional derivative

We can define the derivative of f in the direction of $\mathbf{v}$ originating at the point p

Choose any θ_0 so that



but it did not have any acceleration, it will simply move in a straight line and off the sphere

We define a surface,

$\sigma : U \rightarrow \mathbb{R}^3$ where $U \subset \mathbb{R}^2$ and σ is smooth, and “nice enough” like injective

We will study surfaces by seeing how curves on them are forced to curve

That means, figuring out the acceleration component that keeps the particle moving along the curve on the surface

Multivariable calculus

What is the derivative of

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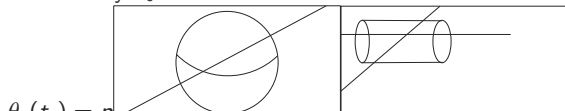
Observations

1. ∇f does not depend on θ
2. $\theta'(t)$ does not depend on f

Directional derivative

We can define the derivative of f in the direction of $\mathbf{v}$ originating at the point p

Choose any θ_0 so that



$$\theta_0(t_0) = p$$

only if it is experiencing an acceleration in the direction of the centre of the sphere

We define a surface,

$\sigma : U \rightarrow \mathbb{R}^3$ where $U \subset \mathbb{R}^2$ and σ is smooth, and “nice enough” like injective

We will study surfaces by seeing how curves on them are forced to curve

That means, figuring out the acceleration component that keeps the particle moving along the curve on the surface

Multivariable calculus

What is the derivative of

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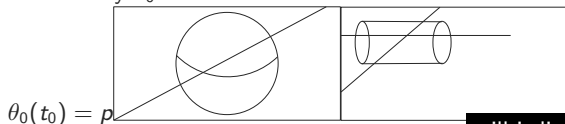
Observations

1. ∇f does not depend on θ
2. $\theta'(t)$ does not depend on f

Directional derivative

We can define the derivative of f in the direction of $\mathbf{v}$ originating at the point p

Choose any θ_0 so that



will it lie on the sphere

We define a surface,

$\sigma : U \rightarrow \mathbb{R}^3$ where $U \subset \mathbb{R}^2$ and σ is smooth, and “nice enough” like injective

We will study surfaces by seeing how curves on them are forced to curve

That means, figuring out the acceleration component that keeps the particle moving along the curve on the surface

Multivariable calculus

What is the derivative of

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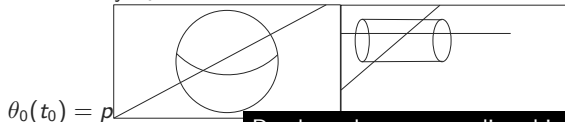
Observations

1. ∇f does not depend on θ
2. $\theta'(t)$ does not depend on f

Directional derivative

We can define the derivative of f in the direction of $\mathbf{v}$ originating at the point p

Choose any θ_0 so that



But how do we generalize this special direction for other surfaces?

We define a surface,

$\sigma : U \rightarrow \mathbb{R}^3$ where $U \subset \mathbb{R}^2$ and σ is smooth, and “nice enough” like injective

We will study surfaces by seeing how curves on them are forced to curve

That means, figuring out the acceleration component that keeps the particle moving along the curve on the surface

Multivariable calculus

What is the derivative of

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}$$

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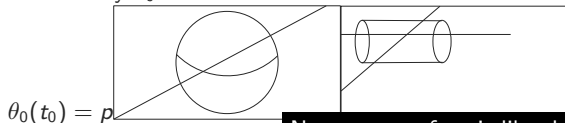
Observations

1. ∇f does not depend on θ
2. $\theta'(t)$ does not depend on f

Directional derivative

We can define the derivative of f in the direction of $\mathbf{v}$ originating at the point p

Choose any θ_0 so that



Not every surface is like the sphere with a special “center”

We define a surface,

$\sigma : U \rightarrow \mathbb{R}^3$ where $U \subset \mathbb{R}^2$ and σ is smooth, and “nice enough” like injective

We will study surfaces by seeing how curves on them are forced to curve

That means, figuring out the acceleration component that keeps the particle moving along the curve on the surface

Multivariable calculus

What is the derivative of

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}$$

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$$\frac{d}{dt} f(\theta(t))$$

Chain rule (a computational device)

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where $\nabla f = (\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y})$ where $\frac{\partial f}{\partial x}$ is obtained by keeping y fixed and taking the derivative with respect to x

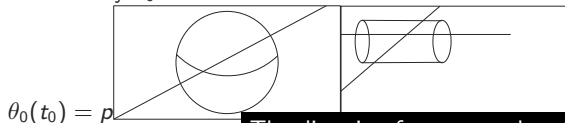
Observations

1. ∇f does not depend on θ
2. $\theta'(t)$ does not depend on f

Directional derivative

We can define the derivative of f in the direction of $\mathbf{v}$ originating at the point p

Choose any θ_0 so that



The direction for a general surface will be called the normal vector

We define a surface,

$\sigma : U \rightarrow \mathbb{R}^3$ where $U \subset \mathbb{R}^2$ and σ is smooth, and “nice enough” like injective

We will study surfaces by seeing how curves on them are forced to curve

That means, figuring out the acceleration component that keeps the particle moving along the curve on the surface

That direction is the so called *normal vector*

Multivariable calculus

What is the derivative of

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}$$

along a parametrization $\theta : (a, b) \rightarrow \mathbb{R}^2$?

$$\frac{d}{dt} f(\theta(t))$$

Chain rule (a computational device)

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where $\nabla f = (\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y})$ where $\frac{\partial f}{\partial x}$ is obtained by keeping y fixed and taking the derivative with respect to x

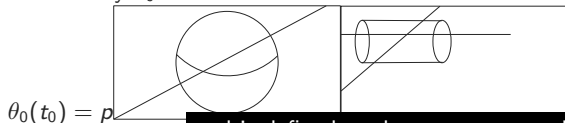
Observations

1. ∇f does not depend on θ
2. $\theta'(t)$ does not depend on f

Directional derivative

We can define the derivative of f in the direction of $\mathbf{v}$ originating at the point p

Choose any θ_0 so that



and is defined as the vector perpendicular to the tangent plane at the surface

We define a surface,

$\sigma : U \rightarrow \mathbb{R}^3$ where $U \subset \mathbb{R}^2$ and σ is smooth, and “nice enough” like injective

We will study surfaces by seeing how curves on them are forced to curve

That means, figuring out the acceleration component that keeps the particle moving along the curve on the surface

That direction is the so called *normal vector* which is the vector perpendicular to the “tangent plane” at that point

Multivariable calculus

What is the derivative of

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}$$

along a parametrization $\theta : (a, b) \rightarrow \mathbb{R}^2$?

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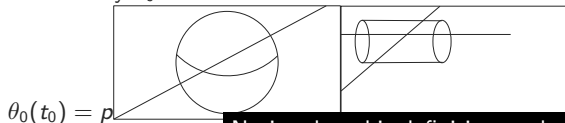
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Directional derivative

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Choose any θ_0 so that



Notice that this definition works fine for the special case of the sphere

We define a surface,

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We will study surfaces by seeing how curves on them are forced to curve

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Multivariable calculus

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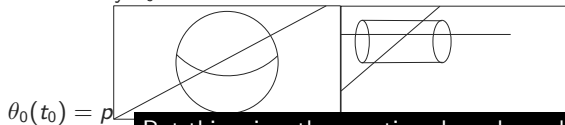
Observations

1. ∇f does not depend on θ
2. $\theta'(t)$ does not depend on f

Directional derivative

We can define the derivative of f in the direction of $\mathbf{v}$ originating at the point p

Choose any θ_0 so that



But this raises the question: how do we know the tangent plane? How do we define it?

We define a surface,

$\sigma : U \rightarrow \mathbb{R}^3$ where $U \subset \mathbb{R}^2$ and σ is smooth, and “nice enough” like injective

We will study surfaces by seeing how curves on them are forced to curve

That means, figuring out the acceleration component that keeps the particle moving along the curve on the surface

That direction is the so called *normal vector* which is the vector perpendicular to the “tangent plane” at that point

How do we define a vector which is tangent to the surface at a point p

Multivariable calculus

What is the derivative of

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}$$

along a parametrization $\theta : (a, b) \rightarrow \mathbb{R}^2$?

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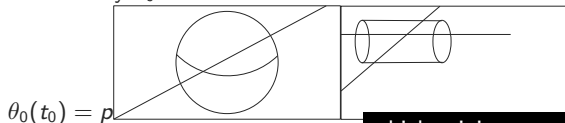
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Directional derivative

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Choose any θ_0 so that



which originate at a point p on the surface

We define a surface,

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We will study surfaces by seeing how curves on them are forced to curve

That means, figuring out the acceleration component that keeps the particle moving along the curve on the surface

That direction is the so called *normal vector* which is the vector perpendicular to the “tangent plane” at that point

How do we define a vector which is tangent to the surface at a point p on the surface?

Multivariable calculus

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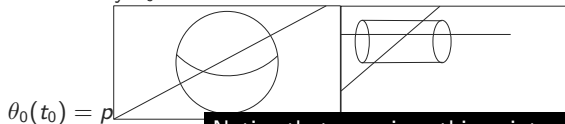
Observations

1. ∇f does not depend on θ
2. $\theta'(t)$ does not depend on f

Directional derivative

We can define the derivative of f in the direction of $\mathbf{v}$ originating at the point p

Choose any θ_0 so that



Notice that σ assigns this point on the surface two coordinates x_0 and y_0 .

We define a surface,

$\sigma : U \rightarrow \mathbb{R}^3$ where $U \subset \mathbb{R}^2$ and σ is smooth, and “nice enough” like injective

We will study surfaces by seeing how curves on them are forced to curve

That means, figuring out the acceleration component that keeps the particle moving along the curve on the surface

That direction is the so called *normal vector* which is the vector perpendicular to the “tangent plane” at that point

How do we define a vector which is tangent to the surface at a point p on the surface?

$$p = \sigma(x_0, y_0)$$

Multivariable calculus

What is the derivative of

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}$$

along a parametrization $\theta : (a, b) \rightarrow \mathbb{R}^2$

$$\frac{d}{dt} f(\theta(t))$$

Chain rule (a computational device)

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where $\nabla f = (\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y})$ where $\frac{\partial f}{\partial x}$ is obtained by keeping y fixed and taking the derivative with respect to x

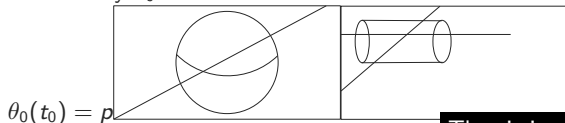
Observations

1. ∇f does not depend on θ
2. $\theta'(t)$ does not depend on f

Directional derivative

We can define the derivative of f in the direction of $\mathbf{v}$ originating at the point p

Choose any θ_0 so that



That is its purpose, afterall!

We define a surface,

$\sigma : U \rightarrow \mathbb{R}^3$ where $U \subset \mathbb{R}^2$ and σ is smooth, and “nice enough” like injective

We will study surfaces by seeing how curves on them are forced to curve

That means, figuring out the acceleration component that keeps the particle moving along the curve on the surface

That direction is the so called *normal vector* which is the vector perpendicular to the “tangent plane” at that point

How do we define a vector which is tangent to the surface at a point p on the surface?

$$p = \sigma(x_0, y_0)$$

Multivariable calculus

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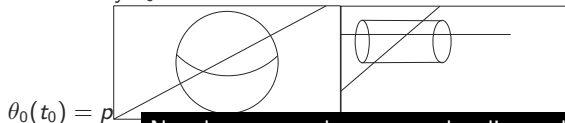
Observations

1. ∇f does not depend on θ
2. $\theta'(t)$ does not depend on f

Directional derivative

We can define the derivative of f in the direction of $\mathbf{v}$ originating at the point p

Choose any θ_0 so that



Now let us see what curves that lie on the part of the surface described by σ look like

We define a surface,

$\sigma : U \rightarrow \mathbb{R}^3$ where $U \subset \mathbb{R}^2$ and σ is smooth, and “nice enough” like injective

We will study surfaces by seeing how curves on them are forced to curve

That means, figuring out the acceleration component that keeps the particle moving along the curve on the surface

That direction is the so called *normal vector* which is the vector perpendicular to the “tangent plane” at that point

How do we define a vector which is tangent to the surface at a point p on the surface?

$$p = \sigma(x_0, y_0)$$

Consider a curve parametrized by $\theta : (a, b) \rightarrow \mathbb{R}^3$ which lies on the surface

Multivariable calculus

What is the derivative of

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}$$

along a parametrization $\theta : (a, b) \rightarrow \mathbb{R}^2$

$$\frac{d}{dt} f(\theta(t))$$

Chain rule (a computational device)

$$\frac{d}{dt} f(\theta(t)) = \nabla f \cdot \theta'(t)$$

where $\nabla f = (\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y})$ where $\frac{\partial f}{\partial x}$ is obtained by keeping y fixed and taking the derivative with respect to x

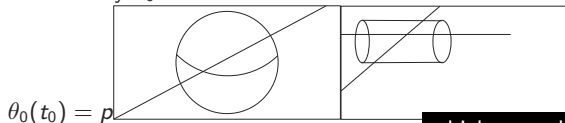
Observations

1. ∇f does not depend on θ
2. $\theta'(t)$ does not depend on f

Directional derivative

We can define the derivative of f in the direction of $\mathbf{v}$ originating at the point p

Choose any θ_0 so that



which pass through the point p

We define a surface,

$\sigma : U \rightarrow \mathbb{R}^3$ where $U \subset \mathbb{R}^2$ and σ is smooth, and “nice enough” like injective

We will study surfaces by seeing how curves on them are forced to curve

That means, figuring out the acceleration component that keeps the particle moving along the curve on the surface

That direction is the so called *normal vector* which is the vector perpendicular to the “tangent plane” at that point

How do we define a vector which is tangent to the surface at a point p on the surface?

$$p = \sigma(x_0, y_0)$$

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Multivariable calculus

What is the derivative of

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}$$

along a parametrization $\theta : (a, b) \rightarrow \mathbb{R}^2$

$$\frac{d}{dt} f(\theta(t))$$

Chain rule (a computational device)

$$\frac{d}{dt} f(\theta(t)) = \nabla f \cdot \theta'(t)$$

where $\nabla f = (\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y})$ where $\frac{\partial f}{\partial x}$ is obtained by keeping y fixed and taking the derivative with respect to x

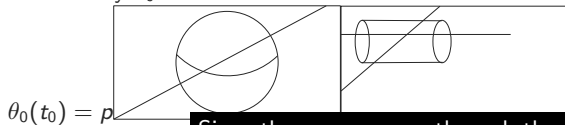
Observations

1. ∇f does not depend on θ
2. $\theta'(t)$ does not depend on f

Directional derivative

We can define the derivative of f in the direction of $\mathbf{v}$ originating at the point p

Choose any θ_0 so that



Since the curve passes through the point p , the point p is itself θ of some t_0

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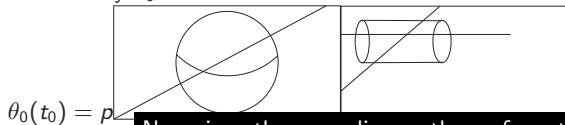
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Directional derivative

We can define the derivative of f in the direction of $\mathbf{v}$ originating at the point p

Choose any θ_0 so that



Now since the curve lies on the surface, the velocity vector is certainly a tangent vector

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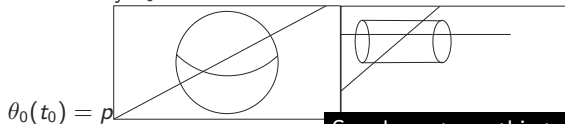
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So why not use this to understand tangent vectors

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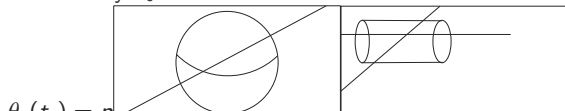
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Choose any θ_0 so that



$$\theta_0(t_0) = p$$

But how do we describe a curve that lies on the part of the surface described by σ

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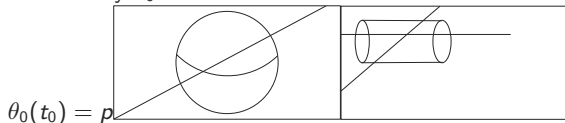
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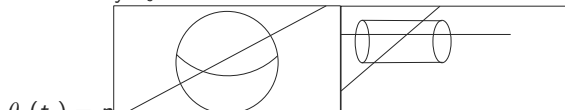
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Choose any θ_0 so that



Since each point lies on the image of σ , each $\theta(t)$ will be of the form $\sigma(x(t), y(t))$

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Since $\theta(t)$ lies on the surface defined by σ , $\theta(t) = \sigma(x(t), y(t))$

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Since $\theta(t)$ lies on the surface defined by σ ,

$$\begin{aligned}\theta(t) &= \sigma(x(t), y(t)) \\ &= (\alpha_1(x(t), y(t)), \alpha_2(x(t), y(t)), \alpha_3(x(t), y(t)))\end{aligned}$$

○

Let us write this out explicitly in terms of the three coordinates

Recall *Chain rule* (a computational device)

$$\frac{d}{dt} f(\theta(t)) = \nabla f \cdot \theta'(t)$$

We define a surface,
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$$\begin{aligned}\theta(t) &= \sigma(x(t), y(t)) \\ &= (\alpha_1(x(t), y(t)), \alpha_2(x(t), y(t)), \alpha_3(x(t), y(t)))\end{aligned}$$

○

Let us now use the chain rule to this expression

Recall *Chain rule* (a computational device)

$$\frac{d}{dt} f(\theta(t)) = \nabla f \cdot \theta'(t)$$

$$\theta'(t_0) =$$

We define a surface,

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Since $\theta(t)$ lies on the surface defined by σ ,

$$\begin{aligned}\theta(t) &= \sigma(x(t), y(t)) \\ &= (\alpha_1(x(t), y(t)), \alpha_2(x(t), y(t)), \alpha_3(x(t), y(t)))\end{aligned}$$

○

Since θ is a composition of functions

Recall *Chain rule* (a computational device)

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We define a surface,

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$$\begin{aligned} \theta(t) &= \sigma(x(t), y(t)) \\ &= (\alpha_1(x(t), y(t)), \alpha_2(x(t), y(t)), \alpha_3(x(t), y(t))) \end{aligned}$$

○

chain rule will give us another way of looking at it

Recall *Chain rule* (a computational device)

$$\frac{d}{dt} f(\theta(t)) = \nabla f \cdot \theta'(t)$$

$$\theta'(t_0) = (x'(t_0) \frac{\partial \alpha_1}{\partial x} +$$

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$$\begin{aligned} \theta(t) &= \sigma(x(t), y(t)) \\ &= (\alpha_1(x(t), y(t)), \alpha_2(x(t), y(t)), \alpha_3(x(t), y(t))) \end{aligned}$$

○

Here is what the first part of the first coordinate looks like

Recall *Chain rule* (a computational device)

$$\frac{d}{dt} f(\theta(t)) = \nabla f \cdot \theta'(t)$$

$$\theta'(t_0) = (x'(t_0) \frac{\partial \alpha_1}{\partial x} +$$

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○

obtained by using the expression highlighted in blue. Make sure you understand why this is the case

Recall *Chain rule* (a computational device)

$$\frac{d}{dt} f(\theta(t)) = \nabla f \cdot \theta'(t)$$

$$\theta'(t_0) = (x'(t_0) \frac{\partial \alpha_1}{\partial x} + y'(t_0) \frac{\partial \alpha_1}{\partial y},$$

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$$\begin{aligned} \theta(t) &= \sigma(x(t), y(t)) \\ &= (\alpha_1(x(t), y(t)), \alpha_2(x(t), y(t)), \alpha_3(x(t), y(t))) \end{aligned}$$

○

and this is what the entire first coordinate looks like

Recall *Chain rule* (a computational device)

$$\frac{d}{dt} f(\theta(t)) = \nabla f \cdot \theta'(t)$$

$$\theta'(t_0) = (x'(t_0) \frac{\partial \alpha_1}{\partial x} + y'(t_0) \frac{\partial \alpha_1}{\partial y}, x'(t_0) \frac{\partial \alpha_2}{\partial x} + y'(t_0) \frac{\partial \alpha_2}{\partial y},$$

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Since $\theta(t)$ lies on the surface defined by σ ,

$$\begin{aligned} \theta(t) &= \sigma(x(t), y(t)) \\ &= (\alpha_1(x(t), y(t)), \alpha_2(x(t), y(t)), \alpha_3(x(t), y(t))) \end{aligned}$$

○

and this is the second coordinate

Recall *Chain rule* (a computational device)

$$\frac{d}{dt} f(\theta(t)) = \nabla f \cdot \theta'(t)$$

$$\theta'(t_0) = \left(x'(t_0) \frac{\partial \alpha_1}{\partial x} + y'(t_0) \frac{\partial \alpha_1}{\partial y}, x'(t_0) \frac{\partial \alpha_2}{\partial x} + y'(t_0) \frac{\partial \alpha_2}{\partial y}, x'(t_0) \frac{\partial \alpha_3}{\partial x} + y'(t_0) \frac{\partial \alpha_3}{\partial y} \right)$$

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○

and finally, this is the third

Recall *Chain rule* (a computational device)

$$\frac{d}{dt} f(\theta(t)) = \nabla f \cdot \theta'(t)$$

$$\begin{aligned} \theta'(t_0) &= (x'(t_0) \frac{\partial \alpha_1}{\partial x} + y'(t_0) \frac{\partial \alpha_1}{\partial y}, x'(t_0) \frac{\partial \alpha_2}{\partial x} + \\ & y'(t_0) \frac{\partial \alpha_2}{\partial y}, x'(t_0) \frac{\partial \alpha_3}{\partial x} + y'(t_0) \frac{\partial \alpha_3}{\partial y}) \\ &= x'(t_0) \frac{\partial \sigma}{\partial x} \Big|_{x=x_0} + y'(t_0) \frac{\partial \sigma}{\partial y} \Big|_{y=y_0} \end{aligned}$$

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○

observe that this can be written in terms of a linear combination of the special vectors $\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$

Recall *Chain rule* (a computational device)

$$\frac{d}{dt} f(\theta(t)) = \nabla f \cdot \theta'(t)$$

$$\begin{aligned} \theta'(t_0) &= (x'(t_0) \frac{\partial \alpha_1}{\partial x} + y'(t_0) \frac{\partial \alpha_1}{\partial y}, x'(t_0) \frac{\partial \alpha_2}{\partial x} + \\ & y'(t_0) \frac{\partial \alpha_2}{\partial y}, x'(t_0) \frac{\partial \alpha_3}{\partial x} + y'(t_0) \frac{\partial \alpha_3}{\partial y}) \\ &= x'(t_0) \frac{\partial \sigma}{\partial x} \big|_{x=x_0} + y'(t_0) \frac{\partial \sigma}{\partial y} \big|_{y=y_0} \end{aligned}$$

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○

Notice that here we will follow our convention of applying the derivative to each coordinate if it is vector valued

Recall *Chain rule* (a computational device)

$$\frac{d}{dt}f(\theta(t)) = \nabla f \cdot \theta'(t)$$

$$\begin{aligned}\theta'(t_0) &= (x'(t_0)\frac{\partial\alpha_1}{\partial x} + y'(t_0)\frac{\partial\alpha_1}{\partial y}, x'(t_0)\frac{\partial\alpha_2}{\partial x} + \\ &y'(t_0)\frac{\partial\alpha_2}{\partial y}, x'(t_0)\frac{\partial\alpha_3}{\partial x} + y'(t_0)\frac{\partial\alpha_3}{\partial y}) \\ &= x'(t_0)\frac{\partial\sigma}{\partial x}|_{x=x_0} + y'(t_0)\frac{\partial\sigma}{\partial y}|_{y=y_0}\end{aligned}$$

Therefore, any tangent vector can be written in terms of the vectors $\frac{\partial\sigma}{\partial x}$ and $\frac{\partial\sigma}{\partial y}$

We define a surface,

$\sigma : U \rightarrow \mathbb{R}^3$ where $U \subset \mathbb{R}^2$ and σ is smooth, and “nice enough” like injective

We will study surfaces by seeing how curves on them are forced to curve

That means, figuring out the acceleration component that keeps the particle moving along the curve on the surface

That direction is the so called *normal vector* which is the vector perpendicular to the “tangent plane” at that point

How do we define a vector which is tangent to the surface at a point p on the surface?

$$p = \sigma(x_0, y_0)$$

Consider a curve parametrized by $\theta : (a, b) \rightarrow \mathbb{R}^3$ which lies on the surface and passes through $p = \theta(t_0)$ then

$\theta'(t_0)$ is an example of a tangent vector

Since $\theta(t)$ lies on the surface defined by σ ,

$$\begin{aligned}\theta(t) &= \sigma(x(t), y(t)) \\ &= (\alpha_1(x(t), y(t)), \alpha_2(x(t), y(t)), \alpha_3(x(t), y(t)))\end{aligned}$$

○

Notice that no matter what θ is, the velocity vector $\theta'(t)$ can always be written in terms of $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$

Recall *Chain rule* (a computational device)

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$$\begin{aligned}\theta'(t_0) &= (x'(t_0) \frac{\partial \alpha_1}{\partial x} + y'(t_0) \frac{\partial \alpha_1}{\partial y}, x'(t_0) \frac{\partial \alpha_2}{\partial x} + \\ & y'(t_0) \frac{\partial \alpha_2}{\partial y}, x'(t_0) \frac{\partial \alpha_3}{\partial x} + y'(t_0) \frac{\partial \alpha_3}{\partial y}) \\ &= x'(t_0) \frac{\partial \sigma}{\partial x} \Big|_{x=x_0} + y'(t_0) \frac{\partial \sigma}{\partial y} \Big|_{y=y_0}\end{aligned}$$

Therefore, any tangent vector can be written in terms of the vectors $\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$

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$$\begin{aligned}\theta(t) &= \sigma(x(t), y(t)) \\ &= (\alpha_1(x(t), y(t)), \alpha_2(x(t), y(t)), \alpha_3(x(t), y(t)))\end{aligned}$$

○

Therefore, the vectors $\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$ are themselves tangent vectors

Recall *Chain rule* (a computational device)

$$\frac{d}{dt} f(\theta(t)) = \nabla f \cdot \theta'(t)$$

$$\begin{aligned} \theta'(t_0) &= (x'(t_0) \frac{\partial \alpha_1}{\partial x} + y'(t_0) \frac{\partial \alpha_1}{\partial y}, x'(t_0) \frac{\partial \alpha_2}{\partial x} + \\ &y'(t_0) \frac{\partial \alpha_2}{\partial y}, x'(t_0) \frac{\partial \alpha_3}{\partial x} + y'(t_0) \frac{\partial \alpha_3}{\partial y}) \\ &= x'(t_0) \frac{\partial \sigma}{\partial x} \Big|_{x=x_0} + y'(t_0) \frac{\partial \sigma}{\partial y} \Big|_{y=y_0} \end{aligned}$$

Therefore, any tangent vector can be written in terms of the vectors $\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$

We can now define the unit normal $\frac{\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y}}{\|\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y}\|}$

○

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and we can then define the unit vector in the direction perpendicular to them by taking the cross product